

Supplementary Material for:

Differentially Private Optimal Transport for Multi-Cloud Intrusion Detection

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This supplement provides: (A) full proof of the utility bound, (B) Byzantine convergence proof, (C) step-by-step RDP composition derivation, (D) per-class detection results, (E) aggregation rule comparison details, and (F) δ sensitivity.

Appendix A: Proof of Theorem 4 (Utility Bound)

Theorem 4 (Restated). *Under bounded feature space and cost matrix assumptions, the Frobenius norm difference between the true and private OT plans satisfies $\|\gamma^* - \tilde{\gamma}^*\|_F \leq O(1/(\sqrt{n}\epsilon))$ with probability $\geq 1 - \delta$.*

Proof. Step 1: Marginal perturbation bound. The Gaussian mechanism adds $\eta \sim \mathcal{N}(0, \sigma^2 I_B)$ with $\sigma^2 = 2\Delta^2 \log(1.25/\delta)/\epsilon^2$ and $\Delta = \sqrt{2}/n$. By sub-Gaussian tail bounds: $\|\eta\|_2 \leq O(\sqrt{B \log(1/\delta)/(n\epsilon)})$ with probability $\geq 1 - \delta$.

Step 2: Sinkhorn Lipschitz continuity. The entropically regularized solution $\gamma_\epsilon^* = \text{diag}(u)K \text{diag}(v)$ where $K = \exp(-C/\epsilon_{reg})$ depends continuously on marginals. The implicit function theorem on the Sinkhorn fixed-point equations yields $\|\partial\gamma^*/\partial a\|_{op} \leq L_K/\epsilon_{reg}$ where $L_K = \exp(2C_{\max}/\epsilon_{reg})$.

Step 3: Composition. Combining via chain rule: $\|\gamma^* - \tilde{\gamma}^*\|_F \leq (L_K/\epsilon_{reg}) \cdot 2\sqrt{B}\|\eta\|_2 \leq O(BL_K\sqrt{\log(1/\delta)/(n\epsilon_{reg}^2\epsilon_{DP})})$. For practical settings ($B = 100$, $\epsilon_{reg} \geq 0.01$, $n > 10^5$), this simplifies to $O(1/(\sqrt{n}\epsilon_{DP}))$. $\square$

Appendix B: Proof of Theorem 5 (Byzantine Convergence)

Theorem 5 (Restated). *Under $q < 1/2$ Byzantine fraction, Algorithm 1 produces $\|\gamma_{global} - \gamma_{true}\|_F \leq O(\sqrt{q/(K(1-q))}) + O(1/\sqrt{n})$.*

Proof. Term 1: Outlier removal. Honest clients' pairwise distances concentrate at $\bar{d} + O(1/\sqrt{n})$. With threshold $\tau = 2 \cdot \text{median}\{\text{med}_k\}$ and honest majority, all honest clients survive while divergent adversaries are flagged.

Term 2: Residual contamination. Surviving adversaries satisfy $W_2(\gamma_k^{adv}, \bar{\gamma}) \leq 2\bar{d}$. With $|\mathcal{B} \cap \hat{\mathcal{H}}| \leq qK$ and $|\hat{\mathcal{H}}| \geq K(1-q)$: Bias $\leq q/(1-q) \cdot O(1/\sqrt{K}) = O(\sqrt{q/(K(1-q))})$.

Term 3: Statistical error. Averaging over honest clients gives $O(1/\sqrt{n})$. $\square$

Appendix C: RDP Composition Derivation

Parameters: $n = 1.6 \times 10^5$, $B = 100$, $\sigma^2 = 5.127 \times 10^{-8}$, $\Delta = \sqrt{2}/n$, $\delta = 5 \times 10^{-4}$, $T = 50$.

Step 1: Per-round RDP. $\rho_1(\alpha) = \alpha\Delta^2/(2\sigma^2) = \alpha \cdot 7.619 \times 10^{-4}$.

Step 2: Composition. $\rho_T(\alpha) = 50 \cdot 7.619 \times 10^{-4}\alpha = 0.03810\alpha$.

Step 3: RDP→DP conversion. $\epsilon(\alpha) = 0.03810\alpha + \log(2000)/(\alpha - 1)$.

Step 4: Optimize. $d\epsilon/d\alpha = 0$ gives $\alpha^* \approx 15.1$, yielding $\epsilon \approx 1.11$.

Step 5: Tightening. Analytical Gaussian calibration (Balle & Wang, 2018) yields $\epsilon = 0.85$.

Comparison: Basic $\epsilon = 4.25$; Advanced $\epsilon = 1.92$; Standard RDP $\epsilon = 1.11$; Optimized RDP $\epsilon = 0.85$.

Appendix D: Per-Class Detection Results

Table 1: Per-class F1-scores (%) on Scenario 3 with $\epsilon = 0.85$.

Attack Type	Freq.	FedAvg	FedKD	WDFT	Ours
Benign	67.3%	85.2	88.4	93.1	96.8
DoS/DDoS	12.1%	71.3	76.8	87.2	93.7
Reconnaissance	5.8%	62.4	69.1	79.3	88.2
Injection	4.2%	54.7	63.5	74.8	86.1
MitM	3.9%	58.1	65.2	76.1	87.5
Malware	2.8%	49.3	58.7	71.4	83.9
CVE-specific	3.9%	39.0	50.1	64.7	78.5
Macro-F1	–	60.0	67.4	78.1	87.8

Appendix E: Aggregation Rule Details

Our Wasserstein-space method uses the W_2 distance between transport plans for outlier scoring, capturing geometric structure that element-wise methods miss. **Krum** selects the plan minimizing sum of distances to nearest neighbors. **Multi-Krum** selects top- m plans by Krum criterion and averages. **Coord. median** applies element-wise median across γ_{ij} entries. **Bulyan** combines Krum selection with trimmed mean. The key advantage of Wasserstein-space aggregation: adversarial plans that induce meaningful misclassification must shift probability mass in geometrically detectable ways, even if individual entries appear normal.

Appendix F: δ Sensitivity

Table 2: Sensitivity to δ at $\epsilon = 0.85$, Scenario 3.

δ_0 (per-round)	δ_{total}	Accuracy (%)	σ^2
10^{-3}	5×10^{-2}	94.4	3.21×10^{-8}
10^{-5}	5×10^{-4}	94.2	5.13×10^{-8}
10^{-7}	5×10^{-6}	94.1	7.04×10^{-8}

Varying δ by four orders of magnitude changes accuracy by only 0.3 points, confirming the trade-off is dominated by ϵ (since $\sigma^2 \propto \log(1/\delta)$).